

Parsa Jafarian

parsa.jafarian@mail.mcgill.ca – linkedin.com/in/parsa-jafarian – github.com/parsajafarian – parsajafarian.github.io

EDUCATION

McGill University

B.Eng in Software Engineering Co-op | GPA: 4.00/4.00

Montreal, QC, Canada

August 2024 – May 2028

EXPERIENCE

Data Engineering Intern

Transport Canada

Dorval, QC, Canada

January 2025 – August 2025

- Migrated 9 out of 24 **ArcGis** data pipelines to **Databricks** using **PySpark** for data manipulation and **Azure Data Factory** for automated data pipelines, contributing to a **\$24K reduction** in annual costs
- Led a team of co-interns to migrate 12 **SAP** reports to **Excel** in under 2 weeks by developing **SQL Server** stored procedures, reducing refresh time by **30%**
- Optimized a **Power BI** dataset by splitting it into **Dataflows**, aggregating data in **Oracle SQL**, and using **DirectQuery**, reducing load time from **2 hours to 20 minutes**
- Used **PowerShell** and **.NET** Power BI REST API libraries to automate the disabling of Power BI refreshes during database failures, ensuring that senior members could resolve issues without interference from ongoing refreshes

Embedded Systems Developer

McGill Formula Electric & McGill Drone Team

Montreal, QC, Canada

September 2024 – January 2025

- Initiated **ROS2** setup for the new Driverless sub-team by building a custom **Docker** Image for containerization, leading to a faster setup for members unfamiliar with Ubuntu or Virtual Machines
- Trained a **YOLO** model to detect black bins on natural terrain using **Kaggle**-sourced data and an **OpenCV**-generated dataset, achieving **98% accuracy** for drone-based fire detection and extinguishing
- Wired an **LCD** to an **Arduino** microcontroller and programmed in **C** to display drone vector data (force, velocity, acceleration) for real-time analytics at the ground station

Web Developer Intern

Tail'ed

Montreal, QC, Canada

July 2024 – August 2024

- Implemented **WebSocket** for real-time communication with **Next.js** server actions, enabling push notifications and reducing server-client data exchange latency from **2 seconds to 100 milliseconds**
- Set up automated unit tests with **Jest** and **GitHub Actions**, reducing backend development time by **40%** and minimizing reliance on slow client-side testing for **MongoDB** data manipulations

SKILLS

Programming Languages: Fluent in Java, Python, TypeScript, SQL; Basic in C, VHDL

Data Science: NumPy, Pandas, Scikit-Learn, Matplotlib, Power BI

Frameworks: React, React Native, Next.js, Express.js, Flask

Dev & Databases: Azure, Docker, Apache Spark, Oracle, MongoDB, Firebase

Languages: Fluent in English, French, Persian; Intermediate in Spanish

PROJECTS

CodeML CN Challenge

github.com/peizheg/code-ml24

- Secured **first place** at **CodeML 2024** by developing a commodity demand prediction model for Canada and integrating Statistics Canada data using **Pandas**
- Trained ML models using **TensorFlow** and **Scikit-Learn**, with the **Random Forest** model demonstrating superior performance, achieving a **93.48%** R² score and a **19.06%** SMAPE score